PPHA 34600 - Problem Set 2 Solutions

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```r
knitr::opts_chunk$set(echo = TRUE, warning = FALSE, message = FALSE)

data <- read.csv("data/ps2_data_spring26.csv")
names(data) <- c(
  "owner_older40",
  "owner_male",
  "num_livestock",
  "years_edu",
  "num_employees",
  "grid_subsidy_amount",
  "grid_connection",
  "firm_revenue"
)
```

Question 1:

PEDAP are interested in answering the following question: What is the average effect of grid electricity connection on firm annual revenues (measured in USD)? To make sure everybody is on the same page, explain to them what the ideal experiment would be for answering this question. Next, use math (the potential outcomes framework) and words to explain what you would estimate and how you would do so. Make sure to be clear about the unit of analysis (i.e., what is "i" here?). Word limit: 250 words

The ideal experiment is a randomized controlled trial (RCT). We would randomly assign grid electricity connections to firms, with some connected to the grid (treatment group) and others remaining unconnected (control group). We would then measure annual firm revenue for both groups and compare the difference as follows:

The unit of analysis refers to firm i.

Let $Y_i(1)$ be the annual revenue of firm i if it is connected to the grid, and let $Y_i(0)$ be the annual revenue of firm i if it is not connected. The treatment effect for firm i is:

$$Y_i(1) - Y_i(0)$$

The parameter we want is the average treatment effect (ATE):

$$ATE = \mathbb{E}[Y_i(1) - Y_i(0)]$$

We know we cannot observe both potential outcomes for the same firm at the same time. Let $D_i = 1$ if the firm is connected and $D_i = 0$ otherwise. If treatment is randomly assigned, then we can estimate the ATE by comparing average revenue across the two groups:

$$\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0]$$

Computing ATE

```
ate_simple <- mean(data$firm_revenue[data$grid_connection == 1]) -
              mean(data$firm_revenue[data$grid_connection == 0])
ate_simple
```

```
## [1] 12992.22
```

Question 2:

PEDAP has some data available, and they believe they can compare firms that are connected to the grid, to those who aren't. They make a dataset available to you, in the form of ps3_data_Spring26.csv. Explain what you would recover if you simply compared connected to non-connected firms. Produce a balance table which displays the differences on observable characteristics between firms that are connected to the grid versus those that are not connected. Interpret this table. Does this table give you more or less faith in the comparison PEDAP would like to make? Word limit: 250 words.

```
# Balance Table

characteristics <- c("owner_older40", "owner_male", "num_livestock", "years_edu", "num_employees", "gri

balance_table <- data.frame()

for (var in characteristics) {
  # Means
  mean_connected <- mean(data[[var]][data$grid_connection == 1], na.rm = TRUE)
  mean_unconnected <- mean(data[[var]][data$grid_connection == 0], na.rm = TRUE)
  difference <- mean_connected - mean_unconnected

  # T-test
  t_test <- t.test(data[[var]][data$grid_connection == 1],
                   data[[var]][data$grid_connection == 0])
  p_value <- t_test$p.value

  # Significance stars
  sig <- ifelse(p_value < 0.01, "***",
                ifelse(p_value < 0.05, "**",
                       ifelse(p_value < 0.10, "*", "")))

  balance_table <- rbind(balance_table, data.frame(
    Variable = var,
    Connected = round(mean_connected, 4),
    Not_Connected = round(mean_unconnected, 4),
    Difference = round(difference, 4),
    P_value = round(p_value, 4),
```

```
    Significant = sig
  ))
}
```

If we simply compare firms that are connected to the grid to firms that are not connected, we recover the difference in average revenues between the two groups. We know this is a naive estimate, and it may not equal the true causal effect of grid connection. The reason is that connected and unconnected firms may already be different in ways that also affect revenue.

The balance table shows several important differences across the two groups. Connected firms are more likely to have older owners, male owners, and owners with more years of education. They also have slightly more employees. The livestock variable looks similar across the two groups and is not statistically different.

This gives us less confidence in the simple comparison PEDAP wants to make. Since the two groups are not well balanced on several observable characteristics, the revenue difference between them is likely also picking up both the effect of grid connection and these existing differences.

Question 3:

(Optional, ungraded question) PEDAP acknowledges that a simple comparison may not work, but they still want an estimate of the impacts of being connected to the grid. PEDAP's CEO recently took a class on impact evaluation and learned about selection-on-observables designs. They have asked you to apply this methodology. Use a regression-based approach to estimate the effect of the grid-connection on firm revenue. Interpret your results. What are the strengths and weaknesses of this approach? How do your results differ from what you find if you instead use the naive estimator? Word limit: 250 words.

```r
# Regression-based Selection-on-Observables Estimate
model_controls <- lm(
  firm_revenue ~ grid_connection +
    owner_older40 +
    owner_male +
    num_livestock +
    years_edu +
    num_employees,
  data = data
)

summary(model_controls)
```

```
##
## Call:
## lm(formula = firm_revenue ~ grid_connection + owner_older40 +
##     owner_male + num_livestock + years_edu + num_employees, data = data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -4670  -1414   -192   1255  11916
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)      4809.75     181.54  26.494  < 2e-16 ***
```

```
## grid_connection 11635.39     101.46 114.679  < 2e-16 ***
## owner_older40    2324.93      99.07  23.467  < 2e-16 ***
## owner_male       1504.85      80.17  18.771  < 2e-16 ***
## num_livestock      38.45      21.51   1.788  0.07397 .
## years_edu         261.73      22.33  11.719  < 2e-16 ***
## num_employees      76.25      25.18   3.029  0.00248 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1965 on 2493 degrees of freedom
## Multiple R-squared:  0.8958, Adjusted R-squared:  0.8956
## F-statistic:  3573 on 6 and 2493 DF,  p-value: < 2.2e-16
```

```r
ate_adjusted <- coef(model_controls)["grid_connection"]
ate_adjusted
```

```
## grid_connection
##        11635.39
```

```r
comparison <- data.frame(
  Estimator = c("Naive Difference in Means", "Regression Adjusted"),
  Estimate = c(ate_simple, ate_adjusted)
)
print(comparison)
```

```
##                                Estimator Estimate
##                 Naive Difference in Means 12992.22
## grid_connection       Regression Adjusted 11635.39
```

Question 4:

Use an exact matching approach to estimate the effect of a grid connection on firm revenue. What variables did you decide to include in the matching procedure (remember you don't have to use all the variables at your disposal)? Interpret your results. What are the strengths and weaknesses of this approach? How do your results differ from what you find if you use the naive estimator? Did you run into the Curse of Dimensionality with this analysis? If yes, describe how it affected your approach. If not, describe how the Curse could have generated problems in this setting. Word limit: 250 words.

```r
library(MatchIt)
library(dplyr)

# Exact matching: covariates chosen for matching wil be education, gender and firm size

m.out <- matchit(
  grid_connection ~ years_edu + owner_male + num_employees,
  data = data,
  method = "exact"
)

# Extract matched dataset
```

```r
matched_data <- match.data(m.out)

# Weighted Matched ATE
ate_weighted <- matched_data %>%
  group_by(subclass) %>%
  summarise(
    diff = mean(firm_revenue[grid_connection == 1]) -
           mean(firm_revenue[grid_connection == 0]),
    n = n()
  ) %>%
  summarise(weighted_ate = weighted.mean(diff, n)) %>%
  pull(weighted_ate)

# Comparison
results <- data.frame(
  Method = c("Naive", "Weighted Matching"),
  Estimate = c(ate_simple, ate_weighted)
)

print(results)
```

```
##               Method Estimate
## 1              Naive 12992.22
## 2 Weighted Matching 12751.43
```

We implemented an exact matching approach by matching firms with and without grid connection on a selected set of pre-treatment covariates (owner education, owner gender, firm size) as matching variables, since these are strongly related to both the likelihood of receiving a grid connection and firm revenue, while keeping the dimensionality manageable.

The main strength of exact matching is that it creates highly comparable treatment and control groups, improving internal validity by eliminating bias from the selected covariates. However, its key weakness is data loss: many observations cannot be matched exactly when multiple covariates are used, reducing sample size and potentially affecting external validity. It also does not adjust for unobserved confounders.

Regarding the Curse of Dimensionality,if more covariates had been included, finding exact matches would become increasingly difficult, potentially leaving very few or no matched pairs.

The resulting estimated effect of grid connection is positive but smaller than the naive estimator of 12,992, suggesting that part of the raw difference was driven by observable differences between firms rather than the treatment itself.

Question 5:

PEDAP has decided they are not interested in using a selection-on-observables approach after your discussions with them. Instead, they mention that they have also heard about a strategy called instrumental variables (IV) and are considering using land gradient as an instrument. They would like you to discuss the plausibility of this instrument and outline its limitations. They would also like you to propose an alternative instrumental variable that could be used to evaluate the effect of grid electricity connection on firm revenues. Why would your proposed instrument be a good one? Do you have any concerns about your ability to estimate the treatment effect using your instrument? If so, explain why; if not, explain why not. Word limit: 250 words.

Land gradient could be a relevant instrument if steeper land makes it harder or more expensive to extend the electricity grid. In that case, firms in steeper areas may be less likely to be connected, so the instrument could predict treatment.

The main problem is the exclusion restriction. A better instrument would be random assignment of a grid connection subsidy. If the subsidy is randomly assigned, it should be unrelated to other firm characteristics, so it is more likely to satisfy the independence condition. It should also be relevant, because a larger subsidy lowers the cost of connecting and should increase the probability that a firm connects to the grid.

The main concern is that this instrument identifies a local average treatment effect, not the effect for all firms. It tells us the effect for firms whose decision to connect changes because of the subsidy.

Question 6:

PEDAP is convinced that this approach is the most feasible given their data limitations. After an internal discussion, they've come back to you with great news! It turns out that one of the many well-meaning development economists at the University of Chicago ran a small pilot program where they randomly offered a grid connection subsidy (covering hookup fees and initial wiring costs) to some firms as part of an attempt to publish a paper in the American Economic Review (it is still unpublished). With this new information, please explain how you would use the subsidy program to estimate the impact of actually connecting to the grid on firm revenues. Use both words and math. Word limit: 250 words.

Since the subsidy was randomly offered, I would use it as an instrument for actual grid connection.

We assume monotonicity, i.e., $D_i(1) \geq D_i(0)$, which rules out the presence of defiers. However, firms may still be always-takers or never-takers, meaning that not all firms offered the subsidy will necessarily connect to the grid.

Let Z_i denote the subsidy amount assigned to firm i, D_i an indicator for whether firm i connects to the grid, and Y_i firm annual revenue. The parameter of interest is the causal effect of grid connection on revenue.

Because D_i is endogenous, we use the subsidy as an instrument. The estimation proceeds via two-stage least squares:

First stage:

$$D_i = \alpha_0 + \alpha_1 Z_i + u_i$$

Second stage:

$$Y_i = \beta_0 + \beta_1 \hat{D}_i + \varepsilon_i$$

Here, β_1 identifies the Local Average Treatment Effect (LATE), which captures the effect of grid connection on firms whose decision to connect is influenced by the subsidy (compliers).

This approach relies on two key assumptions: (i) relevance, the subsidy affects connection, and (iii) exclusion restriction, the subsidy affects revenues only through its impact on grid connection.

Question 7:

PEDAP agree that your approach is a good one. So good, in fact, that they'd like to see it in action! They forgot to tell you that the dataset they provided you will allow you to apply this approach as well. Please report the results of a regression that recovers the impact of grid connection subsidies on the take-up of grid electricity by a firm, using grid_subsidy_amount as the subsidy variable (note that this pilot randomly gave different subsidy amounts to different firms— this is not just a binary variable) and grid_connection as an indicator for having

successfully connected to the national electricity grid. What parameter does this estimate? Interpret your estimate. Word limit: 250 words.

To estimate the impact of the subsidy on grid connection, we run the First Stage Regression:

$$D_i = \alpha + \gamma Z_i + u_i$$

where D_i is an indicator for whether firm i connected to the grid and Z_i is the subsidy amount offered.

```r
first_stage <- lm(grid_connection ~ grid_subsidy_amount, data = data)
summary(first_stage)
```

```
##
## Call:
## lm(formula = grid_connection ~ grid_subsidy_amount, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.40468 -0.24988 -0.17248  0.07752  0.90492
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         0.0176759  0.0196669   0.899    0.369
## grid_subsidy_amount 0.0015480  0.0001185  13.058   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4191 on 2498 degrees of freedom
## Multiple R-squared:  0.0639, Adjusted R-squared:  0.06353
## F-statistic: 170.5 on 1 and 2498 DF,  p-value: < 2.2e-16
```

The estimated coefficient on the subsidy is $\gamma = 0.001548$ ($p < 0.001$), indicating a positive and statistically significant relationship between the subsidy and grid connection. This implies that a one-unit increase in the subsidy amount increases the probability of connecting to the grid by approximately 0.1548 percentage points.

This result provides strong evidence that the subsidy is a relevant instrument, as it significantly affects the take-up of grid electricity. The large F-statistic (170.5) further confirms that the instrument is strong and unlikely to suffer from weak instrument concerns.

Question 8:

PEDAP want you to use the pilot for the next steps of your analysis (they are ignoring any opinion you gave in (5), good or bad – welcome to the real world). They'd first like to learn a little bit more about the pilot itself: they want you to use a regression to estimate the effects of the grid connection subsidies (using grid_subsidy_amount) on firm annual revenues (firm_revenue). What parameter does this regression estimate? Interpret your estimate. Is it useful for policy? Why or why not? Word limit: 250 words.

To estimate the effect of the subsidy on firm revenues, we run the following regression:

$$Y_i = \alpha + \rho Z_i + \varepsilon_i$$

where Y_i is firm annual revenue and Z_i is the subsidy amount.

```r
reduced_form <- lm(firm_revenue ~ grid_subsidy_amount, data = data)

summary(reduced_form)
```

```
##
## Call:
## lm(formula = firm_revenue ~ grid_subsidy_amount, data = data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -9703  -4463  -2058   3938  22844
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)         9388.463    282.808  33.197  < 2e-16 ***
## grid_subsidy_amount   11.504      1.705   6.748 1.85e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6027 on 2498 degrees of freedom
## Multiple R-squared:  0.0179, Adjusted R-squared:  0.01751
## F-statistic: 45.54 on 1 and 2498 DF,  p-value: 1.852e-11
```

```r
#Robust errors
library(sandwich)
library(lmtest)

coeftest(reduced_form, vcov = vcovHC(reduced_form, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                      Estimate Std. Error t value  Pr(>|t|)
## (Intercept)         9388.4633   266.2752 35.2585 < 2.2e-16 ***
## grid_subsidy_amount   11.5036     1.7045  6.7488 1.847e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The estimated coefficient is $\rho = 11.50$ ($p < 0.001$), indicating a positive and statistically significant effect of the subsidy on firm revenues. This implies that a one-unit increase in the subsidy amount increases firm revenue by approximately 11.5 USD.

This parameter corresponds to the reduced-form effect (or intention-to-treat effect), capturing the total impact of the subsidy on revenues. Because the subsidy was randomly assigned, this estimate has a causal interpretation.

However, this estimate does not isolate the effect of actually connecting to the grid, since not all firms offered the subsidy choose to connect. Instead, it reflects both the effect of the subsidy on take-up and the effect of connection on revenues.

This estimate is useful for policy, as it measures the impact of the subsidy program as implemented. However, it does not directly answer the structural question of how much grid connection increases firm revenues. For that, an instrumental variables approach is required.

Question 9:

Finally, PEDAP wants you to use the pilot to estimate the impacts of grid electricity connection on firm revenues. For full transparency, make sure to show all of your analysis steps. PEDAP cares about your standard errors here, so be sure to get them right. Interpret your results: Does grid electricity connection matter for firm revenues? Word limit: 250 words.

To estimate the causal effect of grid connection on firm revenues, we implement a two-stage least squares (2SLS) approach using the subsidy amount as an instrument.

First stage:

$$D_i = \alpha + \gamma Z_i + u_i$$

Second stage:

$$Y_i = \beta + \delta \hat{D}_i + \varepsilon_i$$

where D_i is an indicator for grid connection, Z_i is the subsidy amount, and Y_i is firm revenue.

```r
library(AER)

iv_model <- ivreg(
  firm_revenue ~ grid_connection | grid_subsidy_amount,
  data = data
)

coeftest(iv_model, vcov = vcovHC(iv_model, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                 Estimate Std. Error t value  Pr(>|t|)
## (Intercept)      9257.11     167.33  55.322 < 2.2e-16 ***
## grid_connection  7431.26     604.76  12.288 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The estimated coefficient on grid connection is $\delta = 7431.26$ ($p < 0.001$), indicating that connecting to the grid increases firm revenues by approximately $7,431 USD. This effect is statistically significant and economically large.

This parameter identifies the Local Average Treatment Effect (LATE), capturing the causal effect of grid connection on firms whose decision to connect is influenced by the subsidy (compliers).

Under the assumptions of relevance, independence, and exclusion restriction, this estimate has a causal interpretation. The results suggest that access to reliable electricity substantially improves firm performance, indicating that grid expansion policies could have meaningful economic benefits.

Question 10:

PEDAP like your analysis and are now intrigued by using the subsidy as a policy tool. But they're a bit worried about the quality of their data. While the firm revenue survey data were of extremely high quality, but the pilot research team did not keep great records of the subsidy amounts that were disbursed to firms. The development economist describes two possibilities

and wants your opinion. First, she wants to know if it would be a problem for your analysis — looking at the causal effect of the subsidy on firm revenues — if the subsidy amount data included random noise? Why or why not? If yes, can you use an instrumental variables approach to address this problem? Second, she wants to know if it would be a problem for your analysis if larger subsidy amounts were recorded with error, but smaller subsidy amounts were accurately recorded. If yes, can you use an instrumental variables approach to address this problem? Word limit: 250 words.

If the subsidy amount is measured with random noise, we know this is a classical measurement error problem. This eventually weakens the relationship between the recorded subsidy and the true subsidy, so the estimated effect of the subsidy on firm revenue will usually be biased.

An instrumental variables approach could help. Simply using the noisy subsidy variable itself in an IV framework would not solve the problem.

The second case is more serious. If larger subsidy amounts are measured with more error, while smaller amounts are measured correctly, then the measurement error is not random. In that case, we can think of the error related to the true value of the subsidy, which can create bias in a less predictable way. The problem is harder to fix.

Again, IV could help only if there is another valid instrument for the true subsidy amount that is not affected by this recording problem. Without a separate valid instrument, this type of non-random measurement error would be a serious problem for the analysis and obtaining true or useful results.